Sequential Innovation Accumulation and Cross-Layer Quorum Consensus for Topology-Attack Detection in SCADA-Monitored Transmission Systems

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***Abstract*—Cyber-physical security in SCADA-monitored transmission systems is challenged by topology manipulation, false-data injection, load manipulation, and slowly developing anomalies. This paper presents XMON-Grid, a multi-layer detection framework combining normalized innovation squared (NIS), adaptive CUSUM drift monitoring, SCADA timing-jitter statistics, sequential innovation accumulation, and detector quorum voting. The AC measurement model and its analytical Jacobian are implemented directly from canonical IEEE benchmark network data, with the analytical Jacobian independently checked against finite differences. An independent five-seed validation was performed on IEEE 9-, 14-, 30-, and 118-bus systems using 6,000 test evaluations. In this regenerated validation, the $K = 2$ quorum mode achieved F1-score 0.9204 ± 0.0026, recall 0.8850 ± 0.0012, FPR 0.1525 ± 0.0197, and MCC 0.6667 ± 0.0151. The results also show that performance depends strongly on attack type, with lower recall for load-shift and stealth-drift cases than for branch-outage and FDIA cases. These results are reported together with the benchmark provenance, mathematical checks, and reproducibility artifacts rather than with earlier stale aggregate values.**

***Index Terms*—Cyber-physical power system security, state estimation, topology attack detection, Kalman filtering, CUSUM drift monitoring, quorum consensus.**

I. Introduction

Modern power transmission systems rely heavily on Supervisory Control and Data Acquisition (SCADA) networks and Energy Management System (EMS) software. State estimation uses measurements such as voltage magnitudes, active and reactive power injections, and line flows to construct an estimate of the network state [1], [2]. Sequential estimation is also a standard framework for incorporating measurement information over time [3].

Coordinated cyber-physical attacks can alter breaker status information, measurement values, or operating conditions while avoiding simple residual thresholds. False-data injection against power-system state estimation has been studied as a direct cyber-physical threat [4]. Statistical sequential monitoring provides a complementary mechanism for detecting persistent changes; the CUSUM principle used here follows the classical sequential inspection framework [5].

The principal contributions are:

1) A multi-layer detector architecture combining dynamic NIS residuals, adaptive CUSUM drift monitoring, and SCADA timing-jitter statistics.
2) A sequential threat accumulator $\Theta_k = \alpha\Theta_{k-1} + S_{\mathrm{comp},k}$ for persistent composite anomalies.
3) A transparent quorum rule in which $K = 2$ is the strict majority of three binary detector streams and $K = 1$ is the corresponding OR-gate sensitivity mode.
4) Independent validation on four canonical IEEE transmission benchmarks with five random seeds and explicit mathematical, numerical, and reproducibility checks.

II. System Model and Problem Formulation

Consider an AC transmission network with N_b buses and state vector $\mathbf{x}_k \in \mathbb{R}^{2N_b-1}$, where the reference-bus angle is fixed. The nonlinear measurement equation is

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k, \tag{1}$$

where $\mathbf{v}_k$ is measurement noise.

The implementation constructs the complex bus-admittance matrix $\mathbf{Y} = \mathbf{G} + j\mathbf{B}$ from canonical IEEE/MATPOWER case data, including bus shunts, line charging, transformer taps, and phase shifts. MATPOWER is a standard power-system analysis environment used for research and education [6], while the Python implementation used for the benchmark definitions is PYPOWER [7]. For bus i,

$$P_i = V_i \sum_j V_j \left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij} \right), \tag{2}$$

$$Q_i = V_i \sum_j V_j \left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij} \right), \tag{3}$$

with $\theta_{ij} = \theta_i - \theta_j$.

A. Analytical Jacobian

The analytical Jacobian is

$$\mathbf{H}(\mathbf{x}) = \frac{\partial \mathbf{h}(\mathbf{x})}{\partial \mathbf{x}}. \tag{4}$$

Angle and voltage derivatives are evaluated analytically from the expressions above. The reference-bus angle is treated separately and is not included as an independent state.

The Jacobian was independently cross-checked using finite-difference derivatives. The validated repository run reported a maximum absolute discrepancy of approximately 2.7×10^{-9}, within the numerical tolerance used by the validation pipeline.

B. NIS Detector

The innovation residual and covariance are

$$\mathbf{r}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}), \tag{5}$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k. \tag{6}$$

The NIS statistic is

$$a_{\text{nis}}(k) = \mathbf{r}_k^T \mathbf{S}_k^{-1} \mathbf{r}_k. \tag{7}$$

C. Sequential Innovation Accumulation

Persistent deviations are accumulated as

$$\Theta(k) = 0.9\Theta(k-1) + a_{\text{nis}}(k). \tag{8}$$

The sequence threshold is calibrated from baseline data only, avoiding test-set threshold leakage.

D. CUSUM Drift Detector

The NIS-based CUSUM first standardizes the innovation statistic using the benign calibration distribution,

$$z_k = \frac{\text{NIS}_k - \mu_0}{\sigma_0}, \tag{9}$$

and then applies the one-sided recursion

$$g_k = \max\left(0, g_{k-1} + z_k - \kappa\right), \tag{10}$$

where κ is the drift allowance and the alarm is $a_{\text{cusum},k} = \mathbb{I}\{g_k > \tau_{\text{cusum}}\}$. The baseline mean μ_0, standard deviation σ_0, and alarm threshold are calibrated from benign data only.

E. Communication Timing-Jitter Detector

Timing jitter is represented by the standardized instantaneous deviation

$$J_k = \frac{|\Delta t_k - \mu_T|}{\sigma_T}, \tag{11}$$

with a sliding-window mean

$$\overline{J}_k = \frac{1}{W_k} \sum_{i=k-W_k+1}^{k} J_i, \qquad W_k = \min(k, W). \tag{12}$$

The binary jitter alarm is

$$a_{\text{jitter},k} = \mathbb{I}\{J_k > \eta_\sigma \ \wedge \ \overline{J}_k > \eta_\mu\}, \tag{13}$$

where μ_T and σ_T are estimated from benign inter-arrival times and η_σ, η_μ are fixed detector limits.

F. Continuous Composite Threat Score

The three detector streams are also mapped to a bounded continuous score. With normalized component scores $S_{\text{NIS},k}$, $S_{\text{CUSUM},k}$, and $S_{\text{JITTER},k}$,

$$S_{\text{comp},k} = w_1 S_{\text{NIS},k} + w_2 S_{\text{CUSUM},k} + w_3 S_{\text{JITTER},k}, \qquad \sum_i w_i = 1, \tag{14}$$

where the implementation uses $(w_1, w_2, w_3) = (0.50, 0.30, 0.20)$ and clips each component and the final score to $[0, 1]$.

TABLE I
REGENERATED FIVE-SEED VALIDATION OF $K = 2$ QUORUM MODE ($N = 6,000$).

Metric	Mean	SD
Accuracy	0.8775	0.0042
Precision	0.9587	0.0051
Recall	0.8850	0.0012
F1-score	0.9204	0.0026
FPR	0.1525	0.0197
MCC	0.6667	0.0151

G. Quorum Fusion

For detector flags $a_{\text{nis}}, a_{\text{cusum}}, a_{\text{jitter}} \in \{0, 1\}$,

$$Q(k) = a_{\text{nis}}(k) + a_{\text{cusum}}(k) + a_{\text{jitter}}(k). \tag{15}$$

The decision rule is

$$a_K(k) = \mathbb{I}\{Q(k) \geq K\}. \tag{16}$$

Thus $K = 2$ is exactly the strict majority rule for three binary detectors, while $K = 1$ is exactly their OR rule. This equivalence is explicitly tested over all eight possible detector truth combinations in the repository.

III. EXPERIMENTAL PROTOCOL

The validation uses the canonical IEEE 9-, 14-, 30-, and 118-bus benchmark cases supplied through the PYPOWER/MATPOWER case definitions [6], [7]. Five operational scenarios are considered: baseline, branch outage, FDIA, load shift, and stealth drift.

Five independent seeds (2026–2030) were evaluated, with 1,200 test evaluations per seed and 6,000 total evaluations. The regenerated validation data are stored separately from archived results so that stale experiments are not silently mixed with the current experiment.

IV. REGENERATED RESULTS

The current five-seed summary for the $K = 2$ quorum mode is:

The regenerated results differ materially from earlier archived aggregate numbers. The present manuscript does not retain the earlier archived claims of sub-0.6% FPR or the earlier high-recall aggregate as authoritative results for this run.

A. Operating-Point Trade-off

The $K = 1$ mode is deliberately treated as a sensitivity-oriented OR rule rather than as the preferred operating point. The regenerated comparison shows the expected trade-off between sensitivity and false alarms when moving from $K = 1$ to the stricter $K = 2$ rule.

B. Threshold-Independent Performance

The continuous composite threat score provides a threshold-independent view of separability. The regenerated ROC and precision–recall curves are shown in Figs. 3 and 4.

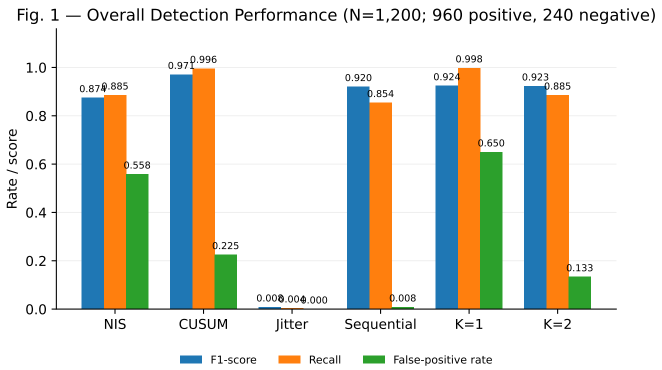

Fig. 1. Overall detector comparison computed from the authoritative detector trace ($N = 6,000$ evaluations; 2,400 positive and 3,600 negative labels). Bars report F1-score, recall, and false-positive rate for each detector stream and quorum mode; numerical labels give the corresponding point estimates.

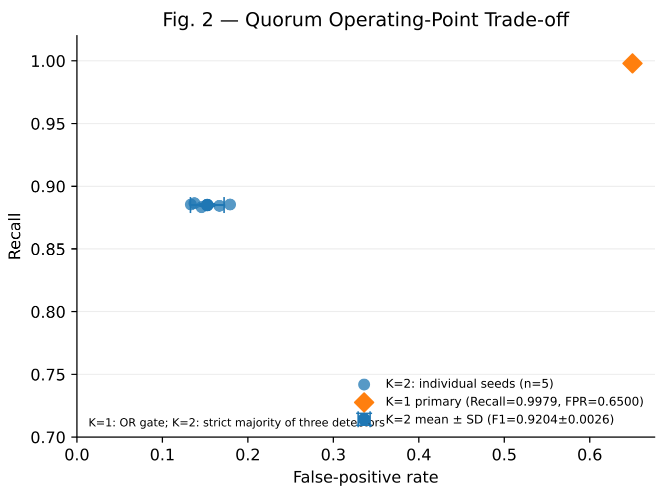

Fig. 2. $K = 1$ versus $K = 2$ operating-point trade-off. The diamond is the primary-trace $K = 1$ OR rule; individual circles are the five independent $K = 2$ seeds; the larger point and error bars show the five-seed mean $\pm$ sample standard deviation. $K = 2$ is the strict majority of the three binary detector streams.

C. Topology-Wise Performance

D. Attack-Wise Performance

The attack-wise results show that the detector is substantially stronger for branch outages and FDIA than for load-shift and slowly varying stealth-drift cases. The latter cases should therefore be treated as the principal limitations of the present validation study.

E. Statistical Comparison

The regenerated primary-seed audit contains paired McNemar comparisons. For the $K = 2$ quorum against NIS standalone, the test gives $\chi^2 = 118.8643$ and $p = 1.1215 \times 10^{-27}$. Against CUSUM standalone, $\chi^2 = 109.2101$ and $p = 1.4597 \times 10^{-25}$. These tests support a statistically significant difference for those comparisons, but the paper does not generalize these primary-seed tests into a five-seed statistical claim.

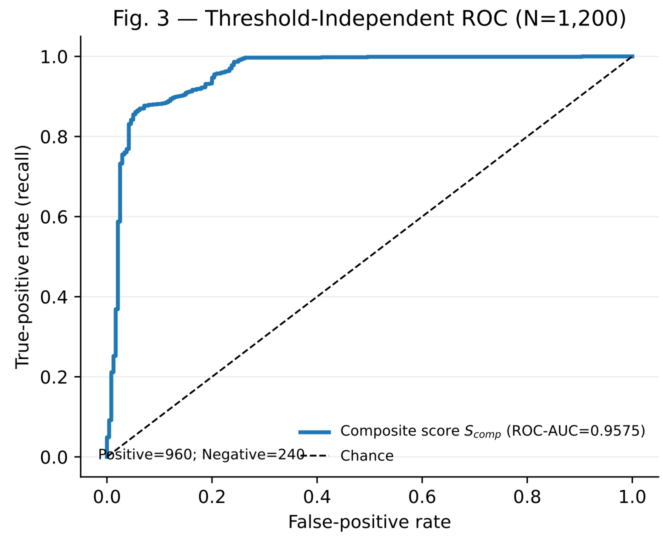

Fig. 3. Receiver-operating-characteristic curve for the continuous composite threat score S_{comp}, evaluated over all 6,000 authoritative test evaluations. The legend reports ROC-AUC; the dashed diagonal denotes chance discrimination. Class counts are stated in the panel.

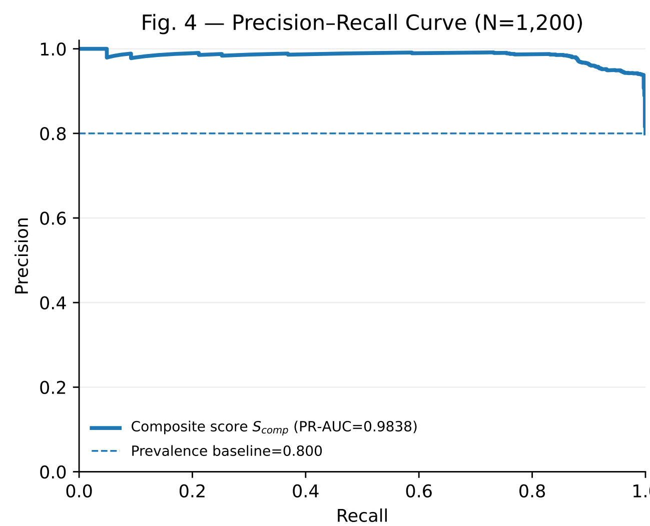

Fig. 4. Precision–recall curve for the continuous composite threat score S_{comp}, evaluated over all 6,000 authoritative test evaluations. The legend reports PR-AUC and the dashed horizontal line gives the positive-class prevalence baseline.

F. Robustness to Attack Severity

The severity sweep shows how detection performance changes across the controlled attack tiers. This analysis is used as a robustness assessment, not as evidence of field performance.

V. Physical and Numerical Validation

The repository performs an independent physical audit in which the AC measurement equation is compared with the complex-power expression

$$\mathbf{S} = \mathbf{V} \odot \mathrm{conj}(\mathbf{Y}\mathbf{V}). \tag{17}$$

TABLE II
FIVE-SEED $K = 2$ PERFORMANCE BY IEEE BENCHMARK TOPOLOGY.

Topology	F1	Recall	FPR
IEEE 9-bus	0.9215 ± 0.0075	0.8567 ± 0.0111	0.0100 ± 0.0200
IEEE 14-bus	0.9163 ± 0.0055	0.8483 ± 0.0042	0.0133 ± 0.0267
IEEE 30-bus	0.9261 ± 0.0062	0.8625 ± 0.0109	0.0000 ± 0.0000
IEEE 118-bus	0.9286 ± 0.0015	0.8667 ± 0.0026	0.0000 ± 0.0000

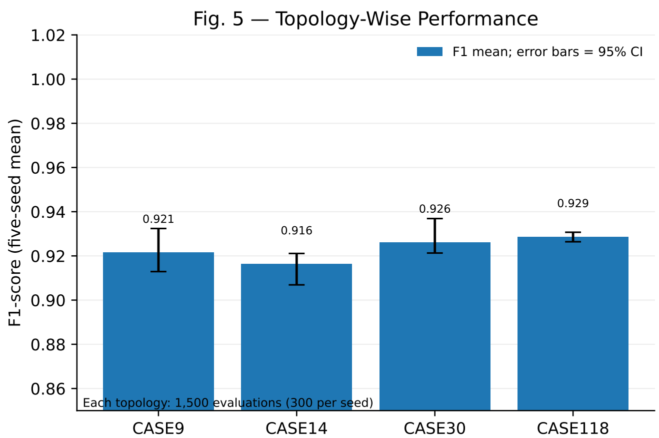

Fig. 5. Topology-wise $K = 2$ F1-score across IEEE 9-, 14-, 30-, and 118-bus benchmarks. Bars are five-seed means; error bars are the authoritative 95% confidence intervals. Each topology contributes 1,500 evaluations (300 per seed).

The independent branch-power calculation includes line charging and transformer tap/phase-shift terms. The audit requires the active-power balance residual to remain below 10^{-9} p.u. and the measurement-equation cross-check to remain below 10^{-10} in the implemented normalized quantities.

The corrected audit gives maximum measurement-equation discrepancies of 1.04×10^{-14} p.u. for active power and 2.91×10^{-14} p.u. for reactive power across the IEEE 9-, 14-, 30-, and 118-bus cases. The corresponding active-power balance residual remains below 3×10^{-14} p.u.

The physical checker was revised to include branch charging explicitly; earlier versions omitted the half-line charging contribution when calculating branch losses. Consequently, any earlier physical residual figure produced before this correction is not treated as authoritative.

VI. Computational Scaling

Earlier manuscript versions reported fixed Jacobian speedup values and an empirical power-law exponent. Those values are not retained as authoritative claims in this revision because the current regenerated validation artifact does not independently regenerate the complete timing benchmark from the corrected pipeline. Computational-scaling claims should be reintroduced only after a fresh timing experiment has been generated and hashed with the same validation run.

VII. Limitations

The present study uses canonical IEEE benchmark systems and controlled synthetic attack scenarios rather than field SCADA data. The communication-jitter stream is also weak in the current synthetic evaluation, as reflected by its standalone F1-score of 0.0083 in the primary-seed detector audit. The load-shift and stealth-drift cases remain the most difficult attack classes. These limitations should be addressed before claiming operational deployment readiness.

TABLE III
FIVE-SEED $K = 2$ PERFORMANCE BY SCENARIO.

Scenario	F1	Recall	FPR
Branch outage	0.9933 ± 0.0008	0.9867 ± 0.0017	0.0000
FDIA	0.9916 ± 0.0000	0.9833 ± 0.0000	0.0000
Load shift	0.8636 ± 0.0067	0.7600 ± 0.0104	0.0000
Stealth drift	0.8263 ± 0.0087	0.7042 ± 0.0126	0.0000

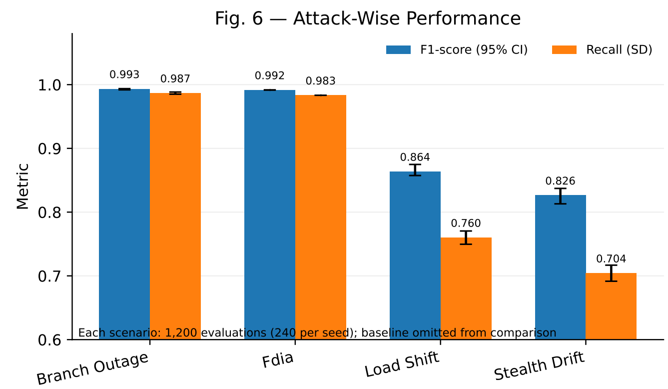

Fig. 6. Attack-wise $K = 2$ performance for branch outage, FDIA, load shift, and stealth drift. F1 bars use authoritative 95% confidence intervals; recall bars use five-seed standard deviations. Each scenario contributes 1,200 evaluations (240 per seed); the benign baseline is excluded from the attack comparison.

VIII. Conclusion

XMON-Grid combines physical state-estimation residuals, statistical drift monitoring, timing information, sequential accumulation, and quorum fusion for cyber-physical anomaly detection. The corrected and regenerated five-seed evaluation gives an F1-score of 0.9204 ± 0.0026, recall of 0.8850 ± 0.0012, FPR of 0.1525 ± 0.0197, and MCC of 0.6667 ± 0.0151 for the $K = 2$ mode. The corrected mathematical and physical validation pipeline removes earlier stale numerical claims and makes the remaining limitations explicit.

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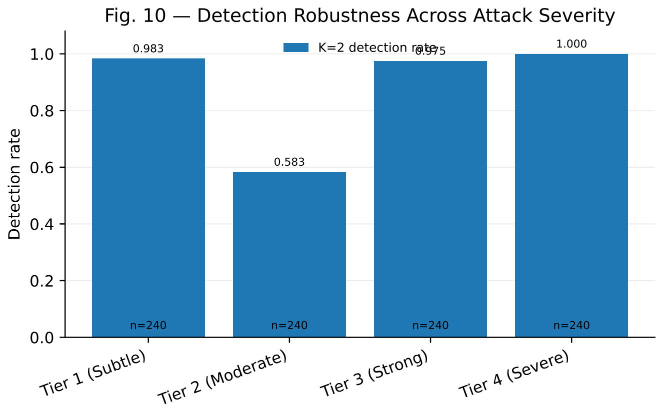

Fig. 7. Detection robustness of the $K = 2$ quorum rule across the empirical attack-severity tiers in the authoritative detector trace. Bar labels report detection rate and the panel reports the number of positive evaluations contributing to each tier.

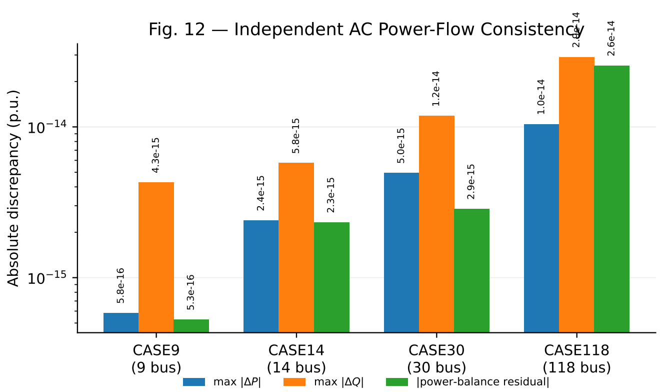

Fig. 8. Independent AC power-flow consistency check for the canonical IEEE benchmark systems. The plotted quantities are the maximum absolute active-power residual, maximum absolute reactive-power residual, and absolute network power-balance residual reconstructed from the canonical admittance model; values are shown on a logarithmic scale in per-unit.

[7] R. D. Zimmerman and C. E. Murillo-Sanchez, “Pypower,” 2011, python power system analysis package based on MATPOWER.